# HIGHER DIRECT IMAGES OF A QUASI-COHERENT MODULE ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM OF ALGEBRAIC SPACES ARE QUASI-COHERENT

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

*Editorial note.* Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

*Allowed standard prerequisites (editorial).*

- Base change of higher direct images along an etale localization, and etale locality of quasi-coherence. Properties of Spaces, labels spaces-properties-lemma-pushforward-etale-base-change-modules (invoked at proof line 351) and spaces-properties-lemma-characterize-quasi-coherent (invoked at proof line 357). Together they reduce the claim to the case where the target $Y$ is an affine scheme.
- An affine etale cover of a quasi-compact algebraic space. Properties of Spaces, label spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof line 365). It provides the surjective etale morphism $g : U \to X$ with $U$ affine, used to run the induction through the Leray spectral sequence of $f \circ g$.
- Quasi-coherence of the zeroth direct image. Morphisms of Spaces, label spaces-morphisms-lemma-pushforward (invoked at proof line 385). It is the base case $n = 0$ of the induction $IH_n$.
- The Leray spectral sequence of a composition. Cohomology on Sites, label sites-cohomology-lemma-relative-Leray (invoked at proof line 396). It relates the higher direct images of $f \circ g$ to those of $f$ and $g$, which is how the induction step is organized.
- Quasi-coherent modules form a weak Serre subcategory of all modules. Properties of Spaces, label spaces-properties-lemma-properties-quasi-coherent (invoked at proof line 408). It is what upgrades the information from the spectral sequence, all of whose other terms in the relevant degrees are known quasi-coherent, into quasi-coherence of $R^{n+1}f_*g_*\mathcal{H}$.

*Date*: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Navigation pointer for \ref{sites-cohomology-lemma-relative-Leray} used inside the counted span (line 396); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03M9: sources/stacks-chow/tags.txt, source lines 4741–4741. Navigation pointer for \ref{spaces-morphisms-lemma-pushforward} used inside the counted span (line 385); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03M0: sources/stacks-chow/tags.txt, source lines 4732–4732. Navigation pointer for \ref{spaces-properties-lemma-characterize-quasi-coherent} used inside the counted span (line 357); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03M1: sources/stacks-chow/tags.txt, source lines 4733–4733. Navigation pointer for \ref{spaces-properties-lemma-properties-quasi-coherent} used inside the counted span (line 408); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03LX: sources/stacks-chow/tags.txt, source lines 4729–4729. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change-modules} used inside the counted span (line 351); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 365); the argument that consumes it is retained in the span.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. If $f$ is quasi-compact and quasi-separated, then $R^i f_*$ transforms quasi-coherent $\mathcal{O}_X$-modules into quasi-coherent $\mathcal{O}_Y$-modules.*

*Proof.* Let $V \to Y$ be an étale morphism where $V$ is an affine scheme. Set $U = V \times_Y X$ and denote $f' : U \to V$ the induced morphism. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. By Properties of Spaces, Lemma 03LX we have $R^i f'_*(\mathcal{F}|_U) = (R^i f_* \mathcal{F})|_V$. Since the property of being a quasi-coherent module is local in the étale topology on $Y$ (see Properties of Spaces, Lemma 03M0) we may replace $Y$ by $V$, i.e., we may assume $Y$ is an affine scheme.

Assume $Y$ is affine. Since $f$ is quasi-compact we see that $X$ is quasi-compact. Thus we may choose an affine scheme $U$ and a surjective étale morphism $g : U \to X$, see Properties of Spaces, Lemma 03H6. Picture

$$\begin{array}{ccc} U & \xrightarrow{g} & X \\ & \searrow^{f \circ g} & \downarrow f \\ & & Y \end{array}$$

The morphism $g : U \to X$ is representable, separated and quasi-compact because $X$ is quasi-separated. Hence the lemma holds for $g$ (by the discussion above the lemma). It also holds for $f \circ g : U \to Y$ (as this is a morphism of affine schemes).

In the situation described in the previous paragraph we will show by induction on $n$ that $IH_n$: for any quasi-coherent sheaf $\mathcal{F}$ on $X$ the sheaves $R^i f \mathcal{F}$ are quasi-coherent for $i \leq n$. The case $n = 0$ follows from Morphisms of Spaces, Lemma 03M9. Assume $IH_n$. In the rest of the proof we show that $IH_{n+1}$ holds.

Let $\mathcal{H}$ be a quasi-coherent $\mathcal{O}_U$-module. Consider the Leray spectral sequence

$$E_2^{p,q} = R^p f_* R^q g_* \mathcal{H} \Rightarrow R^{p+q}(f \circ g)_* \mathcal{H}$$

Cohomology on Sites, Lemma 0734. As $R^q g_* \mathcal{H}$ is quasi-coherent by $IH_n$ all the sheaves $R^p f_* R^q g_* \mathcal{H}$ are quasi-coherent for $p \leq n$. The sheaves $R^{p+q}(f \circ g)_* \mathcal{H}$ are all quasi-coherent (in fact zero for $p + q > 0$ but we do not need this). Looking in degrees $\leq n + 1$ the only module which we do not yet know is quasi-coherent is $E_2^{n+1,0} = R^{n+1} f_* g_* \mathcal{H}$. Moreover, the differentials $d_r^{n+1,0} : E_r^{n+1,0} \to E_r^{n+1+r,1-r}$ are zero as the target is zero. Using that $QCoh(\mathcal{O}_X)$ is a weak Serre subcategory of $Mod(\mathcal{O}_X)$ (Properties of Spaces, Lemma 03M1) it follows that $R^{n+1} f_* g_* \mathcal{H}$ is quasi-coherent (details omitted).

Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Set $\mathcal{H} = g^* \mathcal{F}$. The adjunction mapping $\mathcal{F} \to g_* g^* \mathcal{F} = g_* \mathcal{H}$ is injective as $U \to X$ is surjective étale. Consider the exact sequence

$$0 \to \mathcal{F} \to g_* \mathcal{H} \to \mathcal{G} \to 0$$

where $\mathcal{G}$ is the cokernel of the first map and in particular quasi-coherent. Applying the long exact cohomology sequence we obtain

$$R^n f_* g_* \mathcal{H} \to R^n f_* \mathcal{G} \to R^{n+1} f_* \mathcal{F} \to R^{n+1} f_* g_* \mathcal{H} \to R^{n+1} f_* \mathcal{G}$$

The cokernel of the first arrow is quasi-coherent and we have seen above that $R^{n+1} f_* g_* \mathcal{H}$ is quasi-coherent. Thus $R^{n+1} f_* \mathcal{F}$ has a 2-step filtration where the first step is quasi-coherent and the second a submodule of a quasi-coherent sheaf. Since $\mathcal{F}$ is an arbitrary quasi-coherent $\mathcal{O}_X$-module, this result also holds for $\mathcal{G}$. Thus we can choose an exact sequence $0 \to \mathcal{A} \to R^{n+1} f_* \mathcal{G} \to \mathcal{B}$ with $\mathcal{A}$, $\mathcal{B}$ quasi-coherent $\mathcal{O}_Y$-modules. Then the kernel $\mathcal{K}$ of $R^{n+1} f_* g_* \mathcal{H} \to R^{n+1} f_* \mathcal{G} \to \mathcal{B}$ is quasi-coherent, whereupon we obtain a map $\mathcal{K} \to \mathcal{A}$ whose kernel $\mathcal{K}'$ is quasi-coherent too. Hence $R^{n+1} f_* \mathcal{F}$ sits in an exact sequence

$$R^n f_* g_* \mathcal{H} \to R^n f_* \mathcal{G} \to R^{n+1} f_* \mathcal{F} \to \mathcal{K}' \to 0$$

with all modules quasi-coherent except for possibly $R^{n+1} f_* \mathcal{F}$. We conclude that $R^{n+1} f_* \mathcal{F}$ is quasi-coherent, i.e., $IH_{n+1}$ holds as desired. □